Collatz, but stop at the drop

[subtitle: with three letters, U, D and S of Chapter 8]

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Draft: R_81

1 One language for the drop and the 3-smooth numbers

Papers 1 and 2 study the drop: does every $x > 1$ eventually fall below itself under the map T? Paper 2 reduces it to two open statements, the covering and the width identity (paper 2, Chapter 7). This paper collects what lies around them and is proved, and some of what is measured, in one language: the grid of the numbers $2^j 3^i$. *Proved* means proved in Lean where a name is given; *measured* means computed exactly, on the range given.

One grid

The 3-smooth numbers are the points of the grid, sorted by their binary logarithm $j + i \log_2 3$. The admissible words live on the same grid: the n odd steps of a word of Order n are n points, one per row, a staircase that starts at 3^{n-1} and stays to its left (Chapter 2). The exact links between the two:

- $p_2(n)$, the length of a word of Order n, is the binary length of 3^n, and the sum of the $p_2(k)$ is a term of Mintz's [3] count of the 3-smooth numbers below $2^j 3^i$ (card_smoothBelowAt; paper 2, Chapter 4).
- The record Orders of the block factor $f(n) = 3^n / 2^{p_2(n)}$ decide the blocks of the chain of a word (paper 2), and also the step from a 3-smooth number to the next (smooth_next, smooth_succ; Chapter 2).
- ρ, the constant of $T^L(x) = (3^n x + \rho)/2^L$, is the sum of the staircase's n points, and $d_{23}(n) = 2^{p_2(n)} - 3^n$ is a difference of two 3-smooth numbers. The width identity says that the class id times this difference exceeds the sum of the staircase (ClassDrops, widthIdentity_iff_classDrops).

Whether this language moves either open statement is not known; it is a common language, not yet a tool.

The chapters

Chapter 2 draws the words as staircases and bounds ρ. Chapter 3 takes the largest ρ of each Order as the bound of the finite reduction of the width identity, with its records. Chapter 4 asks whether the congruences that fix a class id carry the drop, and finds that they restate it. Chapter 5 finds the n-th 3-smooth number without listing the others. Chapter 6 is an open statement on neighbouring 3-smooth numbers, with its reductions. Chapter 7 computes the smallest class id of each Order. Chapter 8 plays the game with three letters, and Chapter 9 says what is left out.

Key terms introduced in Chapter 1	
proved	proved in Lean where a Lean name is given, otherwise on paper where the argument is given.
measured	computed exactly, in integers and rational numbers (no floating point), on the range given; not proved.

2 Words as staircases

A word of Order n has n odd steps; let the o-th stand at position k_o, counted from 0. After the whole word, $T^L(x) = (3^n x + \rho)/2^L$ with

$$\rho = \sum_{o=1}^{n} 3^{\,n-o}\, 2^{\,k_o}$$

(paper 2, Chapter 3). Each term is a 3-smooth number $2^j 3^i$, with $i = n - o$ and $j = k_o$. From one odd step to the next the exponent of 3 falls by one and the exponent of 2 rises: the n terms are a *staircase*, one point in each row $i = n-1, \ldots, 0$ of the grid of the 3-smooth numbers.

The slanted grid

Put $2^j 3^i$ at the horizontal position $j + i \log_2 3$, its binary logarithm, in the row i: the *slanted grid*. Then the points of the grid stand from left to right in the order of their values, and a vertical line through a point separates the smaller 3-smooth numbers from the larger ones (the logarithm only places the points; every comparison below is one of integers). In this picture admissibility is a *wall*. In an admissible word the odd step that follows $m \geq 1$ odd steps stands at a position k with $k + 1 \leq p_2(m)$, that is $2^k < 3^m$, so its point $2^k\, 3^{n-1-m}$ is below 3^{n-1}; the first odd step stands at 0, and its point is 3^{n-1} itself. Together with the length $p_2(n)$ this is also enough: a word is admissible exactly when its staircase starts at 3^{n-1}, every later point lies left of 3^{n-1}, and its length is $p_2(n)$ (admissible_iff_steps, admissible_step_le). Figure 1 shows the first word of Order 5.

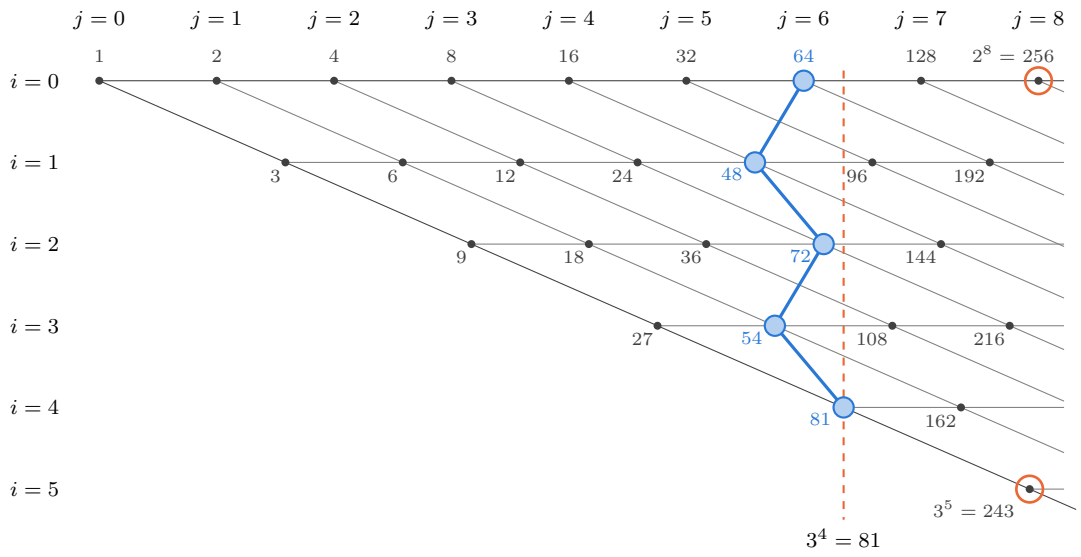


Figure 1: The first word 11011010 of Order 5 as a staircase on the slanted grid: its odd steps at positions $0, 1, 3, 4, 6$ are the points 81, 54, 72, 48, 64, and $\rho = 319$. Each grid point $2^j 3^i$ carries its value. Dashed: the wall at 3^4. Ringed: 3^5 and $2^{p_2(5)} = 2^8$, whose difference 13 is d_{23}.

The bounds on rho

Every point of the staircase is at most 3^{n-1}, so $\rho \leq n\, 3^{n-1}$ (rho_le_of_admissible); Chapter 3 finds the largest ρ, where every point stands as far right as the wall allows. From below, $k_o \geq o - 1$ in every word, so each point is at least $3^{n-o}\, 2^{o-1}$ and $\rho \geq \sum_o 3^{n-o}\, 2^{o-1} = 3^n - 2^n$, with equality exactly for the words that start with all their odd steps (three_pow_le_rho_add, rho_add_eq_three_pow_iff). The staircase of such a word is the diagonal from 3^{n-1} down to 2^{n-1}.

The two ends

The cycle point $x^* = \rho/d_{23}$ has $d_{23}(n) = 2^{p_2(n)} - 3^n$, a difference of two 3-smooth numbers: 3^n at the start of row n, and $2^{p_2(n)}$, the first power of 2 beyond it. A 3-smooth number $2^j 3^i$ lies strictly between them exactly when $1 \le i < n$, $j = p_2(n) - p_2(i)$ and the block factor of i is above that of n (smooth_between_iff). So none lies between them exactly when n is an upper record Order (upperRecord_iff_none_between): in Figure 1, nothing lies between 243 and 256; for $n = 2$, the number $12 = 2^2 \cdot 3$ lies between 9 and 16.

Walking up through the 3-smooth numbers from 3^n to $2^{p_2(n)}$, each step is one of the two of paper 2, Chapter 4: from $2^j 3^i$ the exponent of 3 falls by the last upper record Order $b \le i$, or rises by the last lower record Order a with $p_2(a) \le j + 1$ (smooth_succ). After a power of 3 it is always the step down by b (smooth_succ_of_pow_three); at an upper record Order n that one step, with $b = n$, goes from 3^n straight to $2^{p_2(n)}$.

Over the power series in q

K. G. Monks [4] carries T over to the power series in q over the field with two elements: for A and B with constant term 1, $T_{A,B}(x) = x/q$ when q divides x, and $(Ax + B)/q$ otherwise; each $T_{A,B}$ is conjugate to T by a map that keeps the parities. The staircase goes with it. If x has n odd steps among its first L steps, the o-th at position k_o, then

$$q^L\, T_{A,B}^L(x) = A^n x + B\rho_A, \qquad \rho_A = \sum_{o=1}^{n} A^{\,n-o}\, q^{\,k_o},$$

the formula for ρ with 2 replaced by q and 3 by A, and no signs, as $1 + 1 = 0$ (X_pow_mul_monksT_iterate). For $A = 1$ the map is the shift σ (which deletes the first coefficient) conjugated by B: $T_{1,B}(By) = B\,\sigma(y)$ (monksT_one_mul). Monks lists the series that correspond to $x = 1, \ldots, 20$ for $A = 1 + q$, $B = 1$ and for $A = 1$, $B = 1 + q^2$, and asks for a closed formula. For a start x that reaches 1 after L steps, 1 corresponds to the point $B/(A+q^2)$ of the 2-cycle of $T_{A,B}$, so its series is $(q^L B/(A + q^2) + B\rho_A)/A^n$; for $A = 1$, $B = 1 + q^2$ this is the polynomial $q^L + (1 + q^2)\sum_o q^{k_o}$, of degree L (checked for all 20 rows of both tables). The closed formula is the orbit itself.

Key terms introduced in Chapter 2	
staircase	the n terms $3^{n-o}\,2^{k_o}$ of ρ of a word of Order n, one point in each row of the grid of the 3-smooth numbers.
slanted grid	$2^j 3^i$ drawn at $j + i\log_2 3$ in row i: the points stand in the order of their values.
wall	the vertical line through 3^{n-1}; a word is admissible when every point after the first lies left of it, and its length is $p_2(n)$.

3 The largest rho and the finite reduction

Paper 2 bounds ρ by $n\,3^{n-1}$ for every admissible word of Order n, and so the width identity needs checking only at class ids up to $\tau(n) = n\,3^{n-1}/d_{23}(n)$ (paper 2, Chapter 7). This chapter replaces $n\,3^{n-1}$ by the largest ρ that an admissible word of Order n really has, a sum of 3-smooth numbers, and shows that the finite reduction keeps its shape: its records are again exactly the upper record Orders.

The largest rho

In $\rho = \sum 3^{n-o}\,2^k$ the o-th odd step stands at position k. In an admissible word the odd step that follows m odd steps stands at a position k with $k+1 \le p_2(m)$ (admissible_step_le): before it the level is at least 1. Each term grows with its position, so ρ is largest when every odd step stands as late as it may, at $k = p_2(o-1) - 1$. That gives

$$\rho_{\max}(n) = \sum_{m<n} 3^{\,n-1-m}\,2^{\,p_2(m)-1}, \qquad \rho_{\max}(n+1) = 3\,\rho_{\max}(n) + 2^{\,p_2(n)-1},$$

and $\rho \le \rho_{\max}(n)$ for every admissible word of Order n (rho_le_rhoMax, rhoMax_eq_sum). Every term is a 3-smooth number: in the grid of Chapter 2, each odd step stands as close to the line as admissibility allows. The one admissible word with every odd step at its latest position is the first word of Order n (rank 1), and it is the only word that reaches the bound (rho_eq_rhoMax_iff, rho_lt_rho_firstWord). For $n = 3$ the first word is 11010, with odd steps at 0, 1 and 3, and $\rho_{\max}(3) = 9 + 6 + 8 = 23$, against $n\,3^{n-1} = 27$.

What the bound $n\,3^{n-1}$ gives away, its *slack*, is a sum of the lower deficits $2 \cdot 3^m - 2^{p_2(m)}$:

$$2\left(n\,3^{n-1} - \rho_{\max}(n)\right) = \sum_{m<n} 3^{\,n-1-m}\left(2 \cdot 3^m - 2^{p_2(m)}\right)$$

(rhoMax_slack; for $n = 3$: $0 + 6 + 2 = 8$). Two of its terms compare like $(2 \cdot 3^m - 2^{p_2(m)})\,3^k$ and $(2 \cdot 3^k - 2^{p_2(k)})\,3^m$ (slackTerm_lt_iff), and the term of m is below every earlier one from $k = 1$ on exactly when m is a lower record Order (lowerRecord_iff_slackTerm). So the upper record Orders are where d_{23} is smallest so far, the lower ones where a term of the slack is.

The finite reduction with the largest rho

A class id c of a word of Order n drops at its end when $\rho < d_{23}\,c$, so only the class ids with $d_{23}\,c \le \rho_{\max}(n)$ can fail:

$$\text{the width identity} \iff \text{every class id } c > 1 \text{ with } d_{23}\,c \le \rho_{\max}(n) \text{ drops}$$

(classDrops_iff_small_rhoMax). Per Order, the class ids up to

$$W(n) = \left\lfloor \frac{\rho_{\max}(n)}{d_{23}(n)} \right\rfloor \le \lfloor \tau(n) \rfloor$$

suffice, and a range check of every x with $1 < x \le X$ gives the width identity at every Order with $W(n) \le X$ (classDropsAt_of_widthMax, classDropsAt_of_verifiedMax, widthBoundMax_le_widthBound).

The records

The ratio $\rho_{\max}(n)/d_{23}(n)$, and its floor $W(n)$, are above their values at every earlier Order exactly at the upper record Orders (upperRecord_iff_rmax, upperRecord_iff_widthBoundMax), as for τ. Both factors of

$$\frac{\rho_{\max}(n)}{d_{23}(n)} = \frac{\rho_{\max}(n)/3^n}{1/f(n) - 1}$$

work the same way: $\rho_{\max}(n)/3^n$ grows with n (three_pow_mul_rhoMax_le), and $1/f(n) - 1$ falls as the block factor $f(n)$ of paper 2, Chapter 4, rises. So a later Order with a block factor at least as large has a ratio at least as large (rmax_le_of_le_of_blockFactor_le), and an upper record Order has a ratio above every earlier one.

For the other direction, let n be no upper record Order, a the last upper record Order below it and $b = n - a$. Then p_2 is additive at every split $a + j$ with $1 \le j \le b$, and

$$\rho_{\max}(a+b) = 3^b\,\rho_{\max}(a) + 2^{p_2(a)}\,\rho_{\max}(b) - 3^{\,b-1}\,2^{\,p_2(a)-1}, \qquad d_{23}(a+b) = 3^b\,d_{23}(a) + 2^{p_2(a)}\,d_{23}(b)$$

(rhoMax_add). Without the last term the ratio at $a + b$ would be a mediant of the ratios at a and b, with the weights 3^b and $2^{p_2(a)}$; the term takes it below. With the ratio at b at most that at a (by induction), the ratio at n is below that at a (rmax_add_lt). That the floor W has the same records takes one more step: for two Orders $c < n$ with $f(3) \le f(c) \le f(n)$, the ratio at n is at least 1 above that at c (rmax_add_one_le).

Table 1 puts $W(n)$ beside $\lfloor \tau(n) \rfloor$ at the upper record Orders of paper 2's Table 6. With the last upper record Order a up to n, W of every Order up to n is at most $W(a)$ (widthBoundMax_le_of_upperRecord), so a range check up to $W(a)$ gives the width identity at every Order up to n (classDropsAt_of_verifiedMax_record). For the Orders below 10 781 274 the range check of paper 2 could have stopped at $W(190\,537) = 710\,220\,447\,737$ instead of 984 572 779 224.

n	$p_2(n)$	$\lfloor \tau(n) \rfloor$	$W(n)$
3	5	5	4
17	27	144	108
41	65	1 185	867
306	485	99 729	72 058
15 601	24 727	285 814 986	206 178 386
47 468	75 235	1 447 674 321	1 044 289 613
79 335	125 743	7 216 089 270	5 205 340 379
190 537	301 994	984 572 779 224	710 220 447 737

Table 1: The bound of paper 2 and the bound with the largest ρ, at some upper record Orders (exact).

The bound from Rhin, checked to Order 10^{60}

Paper 2 cites Rhin's [11] lower bound for $2^{p_2(n)} - 3^n$ as a hypothesis (RhinBound) and notes that Lean's kernel checks $3^n \le p_2(n)^2\,(2^{p_2(n)} - 3^n)$ for every $n \le 10^{60}$ (rhinBoundUpTo_one_two). This is a finite check, and no proof uses it; what is of interest is how it avoids the powers 3^n with n near 10^{60}. In product form, with $Q = p_2(n)$, the bound reads $3^n\,(1 + Q^2) \le Q^2\,2^Q$. Between upper record Orders the block factor stays at most its value at the last record a, and p_2 does not fall; so the bound at a gives it at every Order up to the next record (rhinAt_of_record, rhinBoundUpTo_of_records). The upper record Orders are the upper denominators of the walk down the Stern–Brocot tree towards $\log_2 3$ (paper 2, Figure 7), and at each such fraction p/n with $p = p_2(n)$ the check at the fraction computes no power 3^n with $n > 100$: for $n \le 100$ it is exact, and otherwise it comes from a bracket $3^K < 2^b$ with $K = 2^{400}$. Raised to the power K, the bound follows from $b\,n + \lfloor K/\lfloor p^2/2 \rfloor \rfloor + 1 \le p\,K$, a comparison of numbers of at most 181 digits, by $(1 + p^2)^K \le 2^{\lfloor K/\lfloor p^2/2 \rfloor \rfloor + 1}\,p^{2K}$ (rhinOK_spec, succ_pow_le). The walk itself is certified with the same bracket: 649 steps until the sum of its two denominators passes 10^{60}, with 320 upper record Orders among them (rhinWalk_ok, computed by the kernel).

The lower side, and the differences of powers of 2 and 3

The Rhin check bounds the gap $2^{p_2(n)} - 3^n$ above 3^n. The same walk, at its lower fractions, bounds the gap $3^n - 2^{p_2(n)-1}$ below: with $Q = p_2(n)$,

$$2^{\,Q-1}\,(1+Q^2) \le Q^2\,3^n \qquad (1 \le n \le 10^{60})$$

(lowAt_upTo). Between lower record Orders it carries over as the upper side does between upper ones (lowAt_of_records), and the walk checks the lower fraction of every node with the same bracket (lowWalk_ok, computed by the kernel). With both sides, no power of 2 comes close to a power of 3:

$$\min(2^i, 3^j) \le p_2(j)^2\,|2^i - 3^j| \qquad (1 \le j \le 10^{60})$$

(min_le_of_walk), so a number $v = |2^i - 3^j|$ with $j \le 10^{60}$ has $i, j < B$ for every $B \ge 3$ with $(4B^2+1)\,v < 2^B$ (lt_of_walk).

Nothing in these proofs depends on the bound 10^{60} but the walk itself: any walk at $K = 2^m$ that checks every node until its denominators pass N gives both sides up to Order N (rhinBoundUpTo_of_walk, lowAt_of_walk, min_le_of_walks). The bracket has to grow with N: a walk to 10^{100} needs $m = 670$ and 1063 steps, one to 10^{200} needs $m = 1330$ and 5839 steps (computed in integers, with the bracket from $2m+3$ bits as in log2Lo): in each case $K = 2^m$ is just above N^2 (m tried in steps of 10). The kernel has run both walks to 10^{200}, each in four stretches (rhinBoundUpTo_one_two_200, lowAt_upTo_200), from a bracket whose 1330 squarings it checks in fourteen chunks, each from the state before it written out (farLo200_eq, farHi200_eq); so $\min(2^i, 3^j) \le p_2(j)^2\,|2^i - 3^j|$ holds for $1 \le j \le 10^{200}$ (min_le_of_walk_200).

These differences are the terms of OEIS A014121 [6] ($i, j \ge 0$) and A363999 [10] ($i, j \ge 1$). Beyond 10^{200} the bound needs Rhin's, which enters in integers, as the hypothesis RhinInt: when 2^i and 3^n are within a factor 2 ($i \ge 2$), the smaller is at most $i^{14}\,|2^i - 3^n|$. It follows from Rhin's bound by $\min(a,b)\,|\log(a/b)| \le |a-b|$, and assumes less. Under it a finite search decides every v (isPowDiff_iff_searchB). Without any bound, a congruence sieve proves the first 3000 terms of A014121, up to $N = 328\,256\,967\,394\,000\,206\,715$, and the 2891 terms of A363999 up to N (a014121_iff, a363999_iff): modulo $M = 2^8\,3^3\,p_1 \cdots p_{13}$, with the 13 primes $5, 7, 13, \ldots, 1153$ at which the orders of 2 and 3 divide 288 and 1728, the residue of $2^i - 3^j$ for $i \ge 8$, $j \ge 3$ is one of $288 \cdot 1728$, and each of those within N of 0 is the residue of a term; as $2N < M$, the residue gives the number. The cases $i < 8$ or $j < 3$ are checked directly.

Key terms introduced in Chapter 3	
$\boldsymbol{\rho_{\max}(n)}$	the largest ρ of an admissible word of Order n, reached by the first word only.
slack	$n\,3^{n-1} - \rho_{\max}(n)$, a sum of lower deficits.
$\boldsymbol{W(n)}$	$\lfloor \rho_{\max}(n)/d_{23}(n) \rfloor$: the class ids of Order n up to it decide the width identity there.

4 The drop as a congruence

A word w of Order n and length $L = p_2(n)$ has its class id c, its ρ and its $d_{23} = 2^L - 3^n$, and a member x of its class drops at the end of the word exactly when x lies above the cycle point $x^* = \rho/d_{23}$ (paper 2, Chapter 3). The class id is fixed by congruences, one for each piece of the word. This chapter asks whether those congruences carry the drop, and finds that they restate it.

Pieces of a word

For any two words u and v (not necessarily admissible), with n_u odd steps among L_u steps, and so on,

$$\rho(uv) = 3^{n_v}\rho(u) + 2^{L_u}\rho(v)$$

(rho_append), and likewise $2^L - 3^n$ of uv is $3^{n_v}(2^{L_u} - 3^{n_u}) + 2^{L_u}(2^{L_v} - 3^{n_v})$. A number with the steps uv has the steps u, and after them the steps v: so the class id c of uv is congruent to the class id of u modulo 2^{L_u}, and $T^{L_u}(c)$ is congruent to the class id c_v of v modulo 2^{L_v}.

Not block by block

An admissible word splits at its record lows into the blocks of its chain, $w = \boxed{1}B_1 \cdots B_k$ (paper 2, Chapter 4), and each block is an admissible word of lower Order. In one sentence: after the first $\boxed{1}$, each block is a small, local drop that ends below where it began (by a factor between 1/2 and 1, one_lt_deficitRatio, deficitRatio_le_two); the blocks use up the 3/2 of that first $\boxed{1}$, and the word drops when that budget is spent (validOrders_iff_budget; at most one $\boxed{0}$ follows the last block). One could hope that the drop of w follows from the drops of its blocks: that c, carried through the word, passes above the cycle point of each block in turn. It does not, and the cycle point itself shows why. For $w = \boxed{1}\boxed{1}\boxed{0}\boxed{0}$ ($c = 3$, $x^* = 5/7$) the blocks are $\boxed{1}\boxed{0}$ (cycle point 1) and $\boxed{0}$ (cycle point 0). Started at 5/7, the word reaches $11/7 > 1$ before its first block and $10/7 > 0$ before its second: 5/7 passes every block's test, and yet it does not drop, it comes back. So no argument block by block can give the drop.

The bound of a split

What the congruences do give is a bound, the *bound of a split*. Split $w = uv$. As $T^{L_u}(c)$ is a positive number in the class of v, it is at least c_v, the least one; and $T^{L_u}(c) = (3^{n_u}c + \rho(u))/2^{L_u}$, so

$$c \;\geq\; X_u = \frac{2^{L_u}c_v - \rho(u)}{3^{n_u}}.$$

When $X_u > x^*$, the drop at c follows. The bound has a plain meaning. The steps u send the cycle point of uv to the cycle point of the rotation vu, and so

$$X_u > x^*(uv) \iff c_v > x^*(vu)$$

(split_bound_iff, from the identity split_identity in integers). And for the split after the leading $\boxed{1}$, $w = \boxed{1}v$: when $3c+1 \leq 2^L$, the number $T(c) = (3c+1)/2$ is at most 2^{L-1} and has the steps v, so $T(c) = c_v$ (classId_tail_eq_tStep), and the bound of this split is the drop at c itself (split_one_iff). The class ids that could break the width identity are at most $\lfloor n3^{n-1}/d_{23} \rfloor$ (paper 2), and for every Order up to 3000 these satisfy $3c + 1 \leq 2^L$ (computed exactly): there the congruence bound is the width identity in other words.

Measured, on every admissible word of Orders 2 to 12: whenever some split at the end of a block gives $X_u > x^*$, the split after the leading $\boxed{1}$ does. On Orders 2 to 18 only four words have no split that gives the drop: $\boxed{1}\boxed{1}\boxed{0}\boxed{1}\boxed{0}$ and $\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{0}$ (Order 3), $\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{1}\boxed{0}\boxed{0}$ and $\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{0}\boxed{0}$ (Order 5). All four have $3c + 1 > 2^L$, at Orders where d_{23} is small (5 and 13).

The cycle points of the blocks

Measured on Orders 1 to 18: the class id of w lies above the cycle point of every block of its chain. This does not follow from the drop of w: on Orders 2 to 13 the cycle point of w is often below that of a block (for every word of Orders 7 and 12). Nor from the drops of the blocks, as c is not always above the class ids of its blocks. It does follow from a statement per block, measured on Orders 2 to 31 (from Order 14 on by a walk over the prefixes): the class id of the prefix $\boxed{1}\,B_1 \cdots B_i$ lies above the cycle point of B_i, as w and that prefix have the same residue modulo 2^{L_P}, L_P the length of the prefix. But that statement holds the drop of B_i itself whenever T^{L_u} of the prefix's class id is the class id of B_i. So it is no easier than the width identity.

By the index i of the block, the statement gets easier fast. On Orders 2 to 21 the smallest ratio of the class id of $\boxed{1}\,B_1 \cdots B_i$ to the cycle point of B_i is

$$\tfrac{35}{23} \approx 1.52, \quad \tfrac{2015}{227} \approx 8.88, \quad \tfrac{19643}{259} \approx 75.8, \quad \tfrac{13041}{5} \approx 2608, \quad \tfrac{1700937}{5} \approx 340\,187$$

for $i = 1, \ldots, 5$; the first three are reached at Orders 4, 7 and 12, and stay the smallest at every higher Order measured. The cycle point of a block is at most W of its Order, which grows like a power of the Order, while the class id of a longer prefix is fixed by more bits. So the first block is where the statement is tight, and there it contains the width identity: every admissible word B is the first block of $\boxed{1}\,B$, and when c, the class id of $\boxed{1}\,B$, has $3c + 1 \le 2^{1+L_B}$, the class id of B is $T(c) > c$ (classId_tail_eq_tStep); so c above the cycle point of B gives the drop of B. (On Order 14 this holds for 5927 of the 17 637 words.)

Key terms introduced in Chapter 4	
bound of a split	$X_u = (2^{L_u} c_v - \rho(u))/3^{n_u}$ for $w = uv$: the class id c of w is at least X_u.

5 The A003586 program

The n-th 3-smooth number $a(n)$ (OEIS A003586 [5]) can be found without listing the $n-1$ before it. The author's PARI program in that entry does it in two stages, with $\log_2 3$ replaced by a fraction P/Q, and Lean proves the program right, in integers, for every $n \geq 1$. It uses only what Chapters 2 and 3 use: the binary lengths $p_2(i)$ of the powers of 3, and the record Orders.

Record fractions

Call P/Q a *record fraction* when Q is an upper record Order with $P = p_2(Q)$ or a lower one with $P = p_2(Q)-1$ (RecordFrac). Such a fraction gives the binary lengths by floors, $\lfloor iP/Q \rfloor + 1 = p_2(i)$ for $1 \leq i < Q$ (div_record_add_one), and the number of powers of 3 below 2^t as $\lfloor tQ/P \rfloor + 1$ for $1 \leq t < P$ (div_record_swap_add_one).

The first stage: the number of binary digits

The 3-smooth numbers with exactly t binary digits are the $2^{t-p_2(i)}\,3^i$, one for each power of 3 below 2^t. So the count $S(k)$ of the 3-smooth numbers below 2^k is k plus a sum of floors $\lfloor tQ/P \rfloor$, a classical floor sum, computed by Euclidean descent (floorSum_eq, smoothCount_eq); this is Mintz's count of paper 2 at the powers of 2. A bisection (bisect_spec) gives the number B of binary digits of $a(n)$, its rank $r = n - S(B-1)$ among the numbers of B digits, and their count $N = S(B) - S(B-1)$.

The second stage: the rank inside the class

Inside the class of B digits, $2^{B-p_2(i)}\,3^i$ for $i < N$, the numbers are sorted by the block factor $f(i) = 3^i/2^{p_2(i)}$: below one of them lie the 3-smooth numbers below 2^{B-1} and those of the class with a smaller block factor (card_smoothBelowAt_class). With u and v the last lower and the last upper record Order below N (a walk from $(1,1)$, recordWalk_spec), the block factor rises along $i = ku \bmod (u+v)$, $k = 0, 1, \ldots$, skipping $i \geq N$ (card_blockFactor_lt_eq, card_mul_mod_lt): the three-gap theorem. The number of kept i among the first k is a difference of two floor sums (classCount_eq), so a second bisection gives the k of rank r, hence i, and $j = B - 1 - \lfloor iP/Q \rfloor = B - p_2(i)$. The program returns (j, i), and exactly $n-1$ of the 3-smooth numbers lie below $2^j 3^i$ (smoothJl_spec, latticeOrd_smoothJl), for every record fraction P/Q with $P > 2h$, $h = 4 \cdot 2^{\lfloor \log_2(n)/2 \rfloor}$. The program computes floor sums of P, Q and Orders only: it forms no 3-smooth number, not even $a(n)$.

For $n = 51$: $S(11) = 47$, so $B = 12$ and $r = 4$. The class of 12 digits has $N = 8$ members, one for each power of 3 below 2^{12}; $u = 7$ and $v = 5$, and the sweep $k \cdot 7 \bmod 12$ gives $0, 7, 2, (9), 4, (11), 6, 1, (8), 3, (10), 5$, the ones in parentheses skipped: 2048, 2187, 2304, 2592, ... in increasing order. The fourth is $i = 4$, $j = 12 - p_2(4) = 5$: $a(51) = 2^5 \cdot 3^4 = 2592$.

Where the fraction comes from

The PARI program takes P/Q from `bestappr`, which returns a convergent. Every semiconvergent p/n of $\log_2 3$ is a record fraction (recordFrac_of_semiconv), so is every convergent (recordFrac_conv), and with one whose numerator exceeds $2h$ the program returns the exponents of $a(n)$ (latticeOrd_smoothJI_conv). That `bestappr` returns convergents of $\log_2 3$, and not of a nearby number, is a matter of PARI's working precision, outside the proof. Without it, the fraction comes from the Stern–Brocot walk towards $\log_2 3$ in integers: a bracket $2^a \le 3^K < 2^b$ for $K = 2^m$, from fixed-point squaring, decides each mediant p/n by $pK \le an$ (below) or $bn \le pK$ (above), and every step it decides is a step of the true walk, so the walk ends at record fractions (sbCertStep_eq, exists_sbCert). For every large m the walk passes any numerator (exists_lt_walkAt), and the program with the least such m returns the exponents of $a(n)$ for every $n \ge 1$ (latticeOrd_smoothJIFind): no convergent is assumed.

Key terms introduced in Chapter 5	
record fraction	P/Q with Q an upper record Order and $P = p_2(Q)$, or a lower one and $P = p_2(Q) - 1$.
$\boldsymbol{S(k)}$	the number of 3-smooth numbers below 2^k.
A003586	the 3-smooth numbers in the OEIS.

6 The ruler gap

This chapter is an open statement on neighbouring 3-smooth numbers, with its reductions. Papers 1 and 2 do not use it, and nothing in this paper rests on it.

The statement

Draw the 3-smooth numbers on a *plain ruler*: v at distance v from the left, in the row of its number of factors 3. Let $a < b$ be neighbours among them. A second step with the same factor would land at b^2/a; the straight line from a through b, continued over the same number of rows, arrives at $2b-a$, which is before it (line_falls_short). The *ruler gap* says that no 3-smooth number lies strictly between $2b - a$ and b^2/a (RulerGap). For $a = 8$, $b = 9$ the gap is from 10 to $81/8$. Under the ruler gap a 3-smooth number lies before the second step exactly when it lies at or before the end $2b - a$ of the line (rule_iff_of_rulerGap). The same rule with the place where the continued line meets row 0 is false: from 3^{11} the step to $2^8 \cdot 3^6$ is to a neighbour, its line meets row 0 at $197\,996 + 2/5$, and the second step of that factor lands at $2^{16} \cdot 3$, before that place (border_rule_fails).

In exponents and record Orders

The statement is one on exponents only (rulerGap_iff_ordersGap). A step between neighbours that raises the exponent of 2 is a step of an upper record Order b, by the factor $2^{p_2(b)}/3^b$; one that raises the exponent of 3 is a step of a lower record Order a, by $3^a/2^{p_2(a)-1}$ (stepAt_two_three_iff, stepAt_three_two_iff; these are the two steps of Chapter 2), and the ruler gap is a statement on those steps (rulerGap_iff_records). For a step of an upper record Order b, a number in its gap is one more factor $3^v/2^u$ with $u+1 = p_2(v)$, where b and v split at $K = b+v$, $p_2(K)+1 = p_2(b)+p_2(v)$, and

$$1 - f(K) < \big(1 - f(b)\big)^2$$

for the block factor f (gapAt_two_three_iff_deficit); the lower record Orders have their own inequality of this kind. So the ruler gap is the failure of these deficit inequalities wherever the step is due (rulerGap_iff_deficit), and the windows where a step is due end at the next record Order. Over each window the block factor is extreme at one record Order c, so the ruler gap is one inequality per pair of neighbouring record Orders $b < b'$, at the largest upper record Order $c \le b + b'$, and likewise for the lower ones (rulerGap_iff_pairs). In words: the deficit $1 - f$ at such a c is not below the square of the deficit at b. That is a statement on how closely the fractions on the Stern–Brocot walk approach $\log_2 3$, and it is open.

The finite check

Lean's kernel checks the pair inequalities for every pair of neighbouring record Orders up to 10^{60} (upperPairs_upTo, lowerPairs_upTo), with the certified walk of Chapter 3, here at $K = 2^{500}$ (rulerUpWalk_ok, rulerLowWalk_ok). With an upper and a lower record Order between 10^{59} and 10^{60} (exists_records_between), the ruler gap holds for every exponent of 2 and every exponent of 3 up to 10^{59} (rulerGapUpTo_ten_pow). This is a finite check; the ruler gap itself stays open.

Key terms introduced in Chapter 6	
plain ruler	v drawn at distance v, in the row of its number of factors 3.
ruler gap	for neighbouring 3-smooth numbers $a < b$: none lies strictly between $2b - a$ and b^2/a; open (RulerGap).

7 The minimal class ids

The smallest class id of each Order is OEIS A122442 [8], with 5 in the place of the class id 1 of Order 1: $a(n)$ is the smallest $x > 1$ whose drop comes at Order $n-1$, so $2, 5, 3, 11, 7, 39, 287, 231, \ldots$ (paper 2, Chapter 3). The entry defines it by the dropping time, and it is the smallest class id of each Order where the class id drops at the end of its own word, which is checked for every term below.

The scan

The terms are found by a *scan*. A residue $x \bmod 2^W$ fixes the first W steps of x; the residues with no admissible prefix within those W steps are kept, 3 524 586 of them for $W = 28$; for the others the residue fixes the Order. The kept residues are then taken in increasing order of x, round by round, and each x is followed to its drop. As x grows, a term is final as soon as it is found. Each term is checked when it is written: its trajectory reaches its drop exactly at Order $n - 1$, and the drop value is below $a(n)$. The terms $a(1)$ to $a(397)$ are computed so; the last of them is $a(397) = 87\,784\,734\,829\,927$.

Why not a smarter search

The class id of a word is fixed by a congruence modulo 2^L, and that is easy: for one word, $c \equiv -\rho\, 3^{-n} \pmod{2^L}$ (Chapter 4). The hard part is the choice of the word. Two attempts to choose it well, and why they fail:

- *Local moves.* Moving the r-th odd step of a word from position k to k', keeping it admissible, changes ρ by $3^{n-r}\, 2^k\, (2^{k'-k} - 1)$, a local change; but it changes the class id by $-3^{-r}\, 2^k\, (2^{k'-k} - 1)$ modulo 2^L: the bits below k stay, all the bits above can change. A descent by such moves is a restart above the first changed position.

- *A solver.* An SMT encoding (M. Winkler's suggestion: Boolean steps, the levels, the Order, the least x by bisection on a bound) reproduces $a(n)$ and its words for Orders 1 to 25 (measured). Its bit-vector solver fixes the low bits of x first, which is the scan by residues again.

Where the problem sits

In terms of the residues, $c \equiv -\sum_o 3^{-o}\, 2^{k_o} \pmod{2^L}$, with the o-th odd step at position k_o: the smallest class id of an Order asks for a choice of positions, under admissibility, that makes a modular sum small. Problems of this shape can be hard because of the bound. Manders and Adleman [2] showed that deciding whether $x^2 \equiv \alpha \pmod{\beta}$ has a solution with $0 \le x \le \gamma$ is NP-complete (their Theorem 2), also when β is given in factored form; without the bound, square roots modulo a factored β are found in random polynomial time. Here the congruence for one word is linear and its least solution is the residue itself; what is left is the choice of the word, and no reduction from an NP-complete problem to it is known.

There is also a reason why such a reduction is unlikely. Per Order there is one answer, $a(n)$, so with n written in unary the set of triples $(1^n, i, b)$ with bit i of $a(n)$ equal to b has only polynomially many strings of each length: it is sparse. By Mahaney's theorem [1] (Theorem 3.1 and Corollary 3.6 there), if an NP-complete set is many-one reducible in polynomial time to a sparse set, then P = NP. So computing $a(n)$ is not NP-hard in that sense unless P = NP. Neither statement says how hard $a(n)$ is; both say only that the hardness, if any, is not of the usual kind.

Key terms introduced in Chapter 7	
scan	the odd x taken in increasing order over the residues modulo 2^W with no admissible prefix in W steps, each followed to its drop.
A122442	the smallest class id of each Order, in the OEIS.

8 Three letters: the A324245 stop

Papers 1 and 2 play a game with two letters. OEIS A324245 [9], the map of Vaillant and Delarue [12] (paper 2, Chapter 6), has three *letters*. It acts on n; on the odd number $k = 2n+1$ it reads

$$\texttt{U}\colon k \equiv 3 \pmod 4 \mapsto \tfrac{3k+1}{2}, \qquad \texttt{D}\colon k \equiv 1 \pmod 8 \mapsto \tfrac{3k+1}{4}, \qquad \texttt{S}\colon k \equiv 5 \pmod 8 \mapsto \tfrac{k-1}{4}.$$

The steps `U` and `D` go from k to the next odd value of its path under T. The step `S` does not: it replaces k by the smaller odd number $(k-1)/4$, which has the same next odd value, as $3\,(k-1)/4+1 = (3k+1)/4$. Follow k by these steps and *stop* at the first value below k; let M be the number of steps `U` and `D` before the stop. For $k = 7$ the steps are $7 \to 11 \to 17 \to 13 \to 3$, its *letter word* is `UUDS`, and $M = 3$; for $k = 9$ one step `D` goes to 7, and $M = 1$.

The stop and the drop

Before the stop every odd value reached is at least k, and the odd values of the path of k under T are among them, in order. A stop by `D` is a step of that path: its value is the first odd value of the path below k, and M is the Order of the drop of k. A stop by `S` is not: the path of k under T is still at or above k, and from the stop value on it goes the same way, a *merge* before the drop. So M is at most the Order of the drop, with equality exactly when the stop is by `D` (le_order_of_isUdsStop, isUdsStop_of_S; checked for every odd $k < 2^{22}$). The number 7 drops at Order 4, to 5, one odd step after its merge with the path of 3.

The words

Write `U` as 1, `D` as 10 and `S` as 00. The letters multiply k by about 3/2, 3/4 and 1/4: the factors of the bits of a word of paper 2. After a prefix of the letters the value is $P\,k + c$, with P the product of the factors and c fixed by the prefix, so for all large k of a class the stop comes at the first prefix with $P < 1$ (isUdsStop_iff_of_large). It does so for every odd $k < 2^{22}$ (measured); whether it does for every k is open. At the last letter of a class that stops by `D` it follows from the width identity of the word at which the class drops (ClassDrops; paper 2, Chapter 7).

By the product rule the bits of the letters up to the stop are an admissible word of Order M, or one followed by 0. A 1 followed by z zeros splits in exactly one way into letters: `U` and $z/2$ letters `S` when z is even, `D` and $(z-1)/2$ letters `S` when z is odd. The first product below 1 falls at the end of an admissible word or one bit later, and both occur for every admissible word. So for $m \geq 1$ there are $2\,\mathrm{A100982}(m)$ [7] letter words with $M = m$ (paper 2, Chapter 5), $\mathrm{A100982}(m)$ of them fixing k modulo $2^{p_2(m)+1}$ and as many modulo $2^{p_2(m)+2}$ (card_udsStopSet, udsWord_class; counted for $m \leq 16$). For $k = 7$ the bits are 111000, the admissible word 11100 of Order 3 followed by 0. These bits need not be the word of k: in the letters, the steps `S` of an odd value come before its `U` or `D`.

The least odd k with $M = m$, for $m = 1, \ldots, 15$: 3, 11, 7, 39, 95, 155, 191, 127, 603, 359, 239, 159, 639, 283, 743 (measured, $k < 2^{22}$). For $m = 1$ to 4, 7, 8 and 11 to 14 this is the least x that drops at Order $m+1$ (Chapter 7).

Key terms introduced in Chapter 8	
U, D, S	the three steps on odd k: $(3k+1)/2$ for $k \equiv 3 \pmod 4$, $(3k+1)/4$ for $k \equiv 1 \pmod 8$, $(k-1)/4$ for $k \equiv 5 \pmod 8$.
stop	the first value below k reached by the letters.
letter word	the letters of k up to its stop; in bits 1, 10, 00.
M	the number of letters `U` and `D` up to the stop.
merge	a stop by `S`: the stop value joins the path of k under T before its drop.
A100982	the number of admissible words of each Order, in the OEIS.
A324245	the map of the three letters, on $n = (k-1)/2$, in the OEIS.

9 What is left out

The map $3x - 1$

On the negative numbers T is the map $x \mapsto (3x - 1)/2$ for odd x, $x/2$ for even x, with the sign turned. There the drop has a plain answer: $5 \to 7 \to 10 \to 5$ never falls below 5. Compared once with T word by word, it showed the change of sign only, no link between the classes (measured). It says nothing on the drop under T, and this paper leaves it out.

Shares and ratios

Shares, such as how many of the starts of Chapter 8 stop by `D`, and ratios seen on a range are no statements that hold for ever. They are left out.

Other multipliers

Other odd multipliers $q \geq 3$ are left out: paper 2 proves its chain decomposition and sweeps for every odd q, as in M. Winkler's [13], and the lean-files TBracketCountQ, TChainParseQ and TBracketDropQ carry the rest of that chapter over to every odd q.

A The Lean formalization

The Lean names this paper cites are proved in the Lean 4 project AcceleratedCollatz of papers 1 and 2 (paper 2, Appendix A), without `sorry`; open statements and theorems taken from the literature enter as named hypotheses (ClassDrops, RulerGap, RhinBound, RhinInt). Table 2 lists the lean-files by chapter; the names of the PowDiff files are in the namespace PowDiff. Chapter 7 is computed, not formalized.

Ch.	lean-file	content
1, 2	TRho{Bound,Lower}, TSmooth{Between,Next,Place,Succ}, TStaircase, TWidth, TMonksQ	staircases, the bounds on ρ, the 3-smooth numbers in order and between 3^n and $2^{p_2(n)}$, the width identity, ρ over the power series in q
3	TRhin{Bound,Check}, TRho{Max,Slack}, TRhoMax{Bound,Check,Record}	the largest ρ, its slack, the finite reduction and its records, the Rhin check
3	PowDiff{,Far,Rhin,Walk}, PowDiffFar{Bracket,Check,Nodes}, PowDiffFar{Low,Rhin}{1..4}	the lower side to 10^{60} and both sides to 10^{200}, the differences $\lvert 2^i - 3^j \rvert$ (A014121, A363999)
4	TCongruenceSplit	ρ of a concatenation, the bound of a split
5	TBisect, TFloorSum, TRecord{Floor,Walk}, TSmooth{Program,ProgramConv,Rank}, TSternBrocotCert	the A003586 program
6	SmoothRuler, TRuler{Check,Deficit,Pair,Record}	the ruler gap, its reductions and the finite check
8	TUds{,Count,Drop}	the three letters: the stop and the drop, the stop for large k, the classes, the count
9	TBracketCountQ, TChainParseQ, TBracketDropQ	every odd q: the bracket words, the parse, the brackets and the map $qx+1$

Table 2: The lean-files of this paper, by chapter; braces group names as in a shell, so TRuler{Check,Pair} is TRulerCheck and TRulerPair.

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